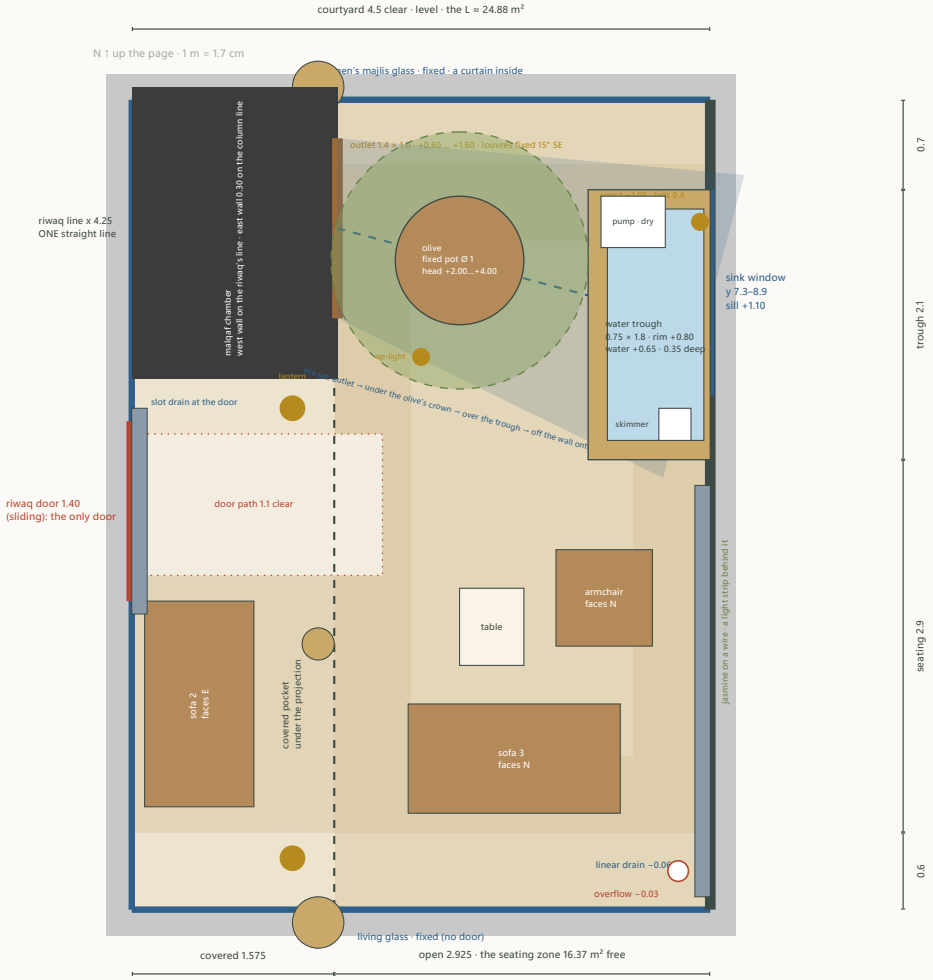
Courtyard layout · H2 revised 11 Nov 2026 · malqaf → olive → water · the straight riwaq · the bigger seating (6 seats) · no TV · Taif / Palm House V10.0

The owner's H2 (`studies/courtyard_haven.typ`, the outlet of `studies/malqaf_outlet_B.typ`) with the owner's three corrections of 11 Nov 2026: the riwaq's line straight, the order outlet → olive → water, the seating extended under the projection. The level floor and the drains of 3 Nov 2026 kept; the TV gone and the core's face straight (10 Nov 2026). Clear dimensions between finished faces; `3d/character.yaml courtyard_layout` follows at the next release (the sink window and the drainage read from it now). Every size assumed; nothing here is for construction; the MEP engineer confirms the jet, the pump and the drains.



1 · The straight riwaq, the level floor and the paths

- **The riwaq's line:** the wall / glass line on x 4.25 runs **straight along its whole length**; the malqaf chamber's west wall stands ON it (the 0.15 jog removed: the chamber 1.55 wide on axes, like the projection above) and its courtyard face is flush with the colonnade line x 5.80 (the east wall 0.30 under the edge beam).
- **Courtyard:** 4.5 × 6.3 m between the finished faces, an L of ≈ 24.88 m², **the whole floor LEVEL** (owner, 3 Nov 2026): the covered strip, the open part and the cleaning strips one surface with gentle falls (page 2); anti-slip porcelain (R11), the field sand, a 0.6 m honey-tone border round the seating zone; no step.
- **Paths kept free:** the **door path** 1.1 m wide (≥ 0.9) from the riwaq's sliding door (1.40, **the only door**: the living's glass stays fixed) straight east into the open part (to x 6.3); the living's glass (0.75 m); the women's majlis glass (0.7 m to the trough and the pot, a **curtain inside the majlis**); 0.45 m behind the armchair at the kitchen wall; 0.7 m between the armchair and the trough's rim to clean it.

2 · Six seats with backs, facing the olive and the water · the bigger zone

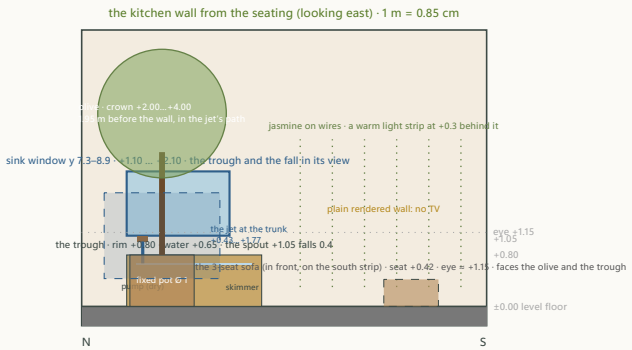
- **The seating zone** (owner, 11 Nov 2026): the open part 2.925 × 5.2 = 15.21 m² less the trough 1.99 m², **plus the covered pocket under the first-floor projection** 1.575 × 2 = 3.15 m² (x 4.35–5.925, south of the door path): **16.37 m² free** (13.2 before); the pocket the shaded summer seat.
- **Modern movable pieces** (the owner chooses them), **every seat with a back, no poufs or stools**: a **3-seat sofa** 1.65 × 0.85 on the south strip **facing north: the olive ahead-left, the trough ahead-right**; a **2-seat sofa** 1.6 × 0.85 in the covered pocket with its back at the riwaq's glass **facing east: the olive, the water beyond it**; an **armchair** by the kitchen wall **facing north onto the trough** with the olive at its left; the low **table** 0.5 × 0.6 in the middle: **6 seats**, every seat seeing the olive and the water; the farthest seat-to-seat ≈ 3.2 m. Outdoor fabrics; the cushions stored in the hospitality kitchen.
- **The olive** (owner, 11 Nov 2026): a 2–3 m standard in a **fixed pot Ø 1 × 0.8 m** on pot feet **in the jet's path between the outlet and the water** (x 6.4–7.4, y 7.35–8.35 its trunk on the jet's axis 0.95 m from the core's face, 0 m from the trough's rim): the **crown +2.00…+4.00** above the jet's top (+1.77 at the trunk) and above the sink's eye line, pruned off the core's face; from the sink the trunk stands 1.95 m beyond the trough, the crown's underside in the window's top; ≈ 1.2 t at one spot (a note for the engineer); the saucer piped to the kitchen-wall drain; a drip ring on a timer; the leaves blow east onto the water, to the skimmer.

3 · The breeze: outlet → olive → water · the trough under the sink window

- **The outlet** (option B, adopted): 1.4 × 1.0, raised to **+0.60…+1.60** and centred in the first arcade bay (y 6.9–8.3), the reveals splayed and the **timber louvres fixed 15° to the south-east** (owner, 11 Nov 2026: re-aimed from 35° so that the jet runs over the olive's pot onto the trough; the damper D2 behind them unchanged): at the summer-evening mean wind ≈ 1.79 m³/s leave at ≈ 2.1 m/s; the jet's axis passes through the olive's trunk **under its crown** (the pot's rim and the trunk take the jet's lowest 0.35 m) and meets the kitchen wall at y 8.38, the trough's middle: **100 % of the jet's width at the wall is over the water** (y 7.85–8.95 within the trough's 7.3–9.4); at the trough's near rim the jet spans +0.24…+1.96 (68 % of its depth over the water); off the wall it spreads south and settles on the three seats at ≈ 0.6 m/s (±30 %, the MEP engineer confirms), a little cooler for the water; nobody faces the jet within 3 m.
- **The trough:** 0.95 × 2.1 outside (walls 0.15) **on the kitchen wall in front of the sink window** (y 7.3–9.4 over the window's 7.3–8.9), the water 0.75 × 1.8 and 0.35 deep, **rim +0.80** (below the sill +1.10: the water and the fall seen from the sink; the sight line reaches +0.42 at the far rim), a **wall spout at +1.05 falling 0.4 m** at its north end under the window: a calm, audible fall; the surface flow runs south to the **skimmer basket** (the olive's leaves); the pump dry under the spout; tanked, its drain and overflow to the rain system.
- **Light:** two lanterns under the soffit edge, a light under the spout, the olive up-lit, a 2700 K strip behind the jasmine; one switch by the riwaq door and a dimmer; nothing shining into the rooms. **Planting:** the olive, the jasmine on a wire (y 10.0–12.6), a laurel in a smaller pot out of the jet.

Study checks passed (10)

4 · The kitchen wall from the seating (the trough under the window, the olive before it, the jasmine; no TV) · 5 · Drainage on a level floor · 6 · Dimensions



4 · The kitchen wall: the trough under the window, the spout, the jasmine

- **The trough** 0.95 × 2.1 against the wall face in front of the sink window (y 7.3–9.4), rim +0.80: a **spout** (a short copper or bronze lip, Ø 25) at +1.05 at the north end, under the window, gives a calm fall of 0.4 m heard from every seat (≈ 45–50 dB(A) at 2 m, assumed) and keeps the surface moving; the flow runs south to the **skimmer basket** (emptied by hand: the olive's leaves); the pump (≈ 40 W, a float switch) dry under the spout, a hatch in the rim; the trough tanked, its drain and overflow into the wall-foot drain's pipe (the rain system).
- **The sink's view:** the rim and the water right under the sill (+1.10), the fall at the window's north end; the olive's trunk 1.95 m beyond, its crown from +2.00 in the window's top (above the eye at the sink, ≈ +1.6): the window not blocked, the seating and the riwaq seen past the trunk.
- **The wall** south of the trough plain render: the **jasmine** on stainless wires (y 10.0–12.6) with a warm light strip behind it at +0.3; the armchair 0.45 m off the wall.

5 · Drainage on a level floor

- **Falls:** the inside floors ±0.00; the courtyard finish at the three glass lines and the riwaq door at **−0.02** (a 15 mm threshold, R-08), falling **1 % away from the glass** across the floor to the kitchen wall foot at **−0.06**: no step, a 45 mm difference over the width that nobody feels; the covered pocket falls the same way.
- **Linear drains:** a slot drain with a silt basket **along the kitchen wall foot** south of the trough (y 9.6–12.8 the trough's and the pot's pipes join it) takes the whole floor; a short slot drain **at the riwaq door's threshold** (wind-driven rain at the door, under the door path); the olive's saucer pipe (Ø 32 in the build-up) runs east to the wall-foot drain's pipe. **Overflow route:** a second gully at the south-east corner (rim −0.03), its own pipe east, so a blocked drain floods the courtyard 3 cm, never the rooms.
- **Pipes:** east under the kitchen floor (invert −0.55, 1 % fall) to the **east RAIN chamber** and the soakaway (`studies/plumbing.typ`): **never into the foul system**. **Waterproofing:** one membrane over the courtyard slab (lowered 0.05 for the build-up), up 0.15 behind the glass plinths and the kitchen wall base; the trough tanked separately.
- **Engineer / MEP to confirm:** the drains and pipes, the rain chamber, the pump's circuit, the pot's point load (≈ 1.2 t, mid-slab), the lighting and two sockets.

6 · Dimensions (clear, between finished faces)

Item	m	Note
Courtyard clear, level	4.5 × 6.3	the L ≈ 24.88 m² clear (27.19 on axes); no step
Malqaf chamber	1.6 × 2.27	x 4.25–5.80 on axes: the west wall on the riwaq's line, the east wall 0.30 on the column line
Seating zone (level)	16.37 m² free	open 15.21 − trough 1.99 + covered pocket 3.15
Door path	1.1 wide	from the sliding door east to x 6.3 kept free (≥ 0.9)
Furniture	6 seats	a 3-seat sofa, a 2-seat sofa, an armchair (every seat with a back), a low table; movable
The olive's pot	Ø 1 × 0.8	x 6.4–7.4, y 7.35–8.35, on the jet's axis; head +2.00…+4.00 ≈ 1.2 t
Cleaning strips	0.75 S · 0.7 N · 0.45 E · 0.7 at the trough	living glass / majlis glass / kitchen wall / the trough's rim
Trough	0.95 × 2.1 × 0.35	under the sink window, y 7.3–9.4 rim +0.80, water +0.65, spout +1.05 walls 0.15
Malqaf outlet (H2 / B)	1.4 × 1	y 6.9–8.3, +0.60…+1.60 louvres fixed 15° SE; 100 % of the jet's width over the water at the wall
TV	none	removed with its surround (owner, 10 Nov 2026)
Levels	−0.02 / −0.06	glass lines / wall foot; overflow −0.03 invert −0.55

What changed and where

- **3 Nov 2026:** the level floor. **10 Nov 2026:** no TV; the core's face straight; **H2** (`studies/malqaf_outlet_B.typ`, `studies/air_system.typ`): the outlet raised, the trough and the olive's pot in place of the planter and the tree. **11 Nov 2026 (owner):** the riwaq's line straight (the chamber's west wall on x 4.25); the order outlet → olive → water (the trough under the window, the olive in the jet, the louvres 15°); the seating extended under the projection (six seats, a 1.1 m door path).
- **Next release:** `3d/character.yaml`, the 3D, `3d/params.yaml`, the DXFs, `studies/plumbing.typ`, `PROJECT_V10.md`, `DESIGN_LANGUAGE.md`.